AP CALCULUS AB

Unit 1 • Worksheet 1-1A • Answer key

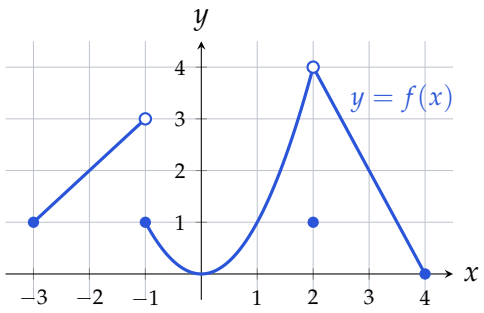
Worksheet 1-1A • Limits to a Point on a Graph

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Questions 1 to 5 refer to the graph of f.



Question 1

NO CALCULATOR

What is $\lim_{x \to -1^-} f(x)$?

(A) 3

(B) 1

(C) 2

(D) It does not exist.

Solution

Answer: (A). From the left the segment approaches the open circle at height 3.

Question 2

NO CALCULATOR

What is $\lim_{x \to 2} f(x)$?

(A) 1

(B) 2

(C) 4

(D) It does not exist.

Solution

Answer: (C). Both pieces approach $(2, 4)$.

Question 3

NO CALCULATOR

For which value of x in $(-3, 4)$ does $\lim f(x)$ fail to exist?

(A) $x = 2$
(B) $x = 0$
(C) $x = 3$
(D) $x = -1$

Solution
Answer: (D). One-sided limits 3 and 1.

Question 4

NO CALCULATOR

What is $\lim_{x \to 1} \big(f(x) + 2x\big)$?

(A) 1
(B) 3
(C) 2
(D) 4

Solution
Answer: (B). $1 + 2 = 3$.

Question 5

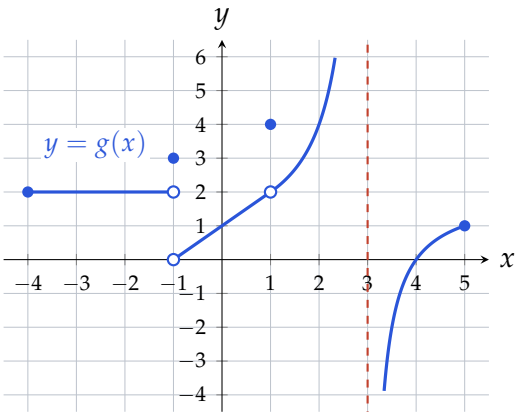
NO CALCULATOR

Which statement is true?

(A) $\lim_{x \to 2} f(x) = f(2)$
(B) $\lim_{x \to -1^+} f(x) = f(-1)$
(C) $\lim_{x \to -1} f(x) = 1$
(D) $f(2) = 4$

Solution
Answer: (B). $\lim_{x \to -1^+} f(x) = 1 = f(-1)$. The limit at 2 is 4 but $f(2) = 1$.

Question 6 refers to the graph of g.



Question 6 (Free response)

NO CALCULATOR

Use the graph of g.

(a) Find $\lim_{x\to -1^-} g(x)$, $\lim_{x\to -1^+} g(x)$ and $g(-1)$. Does $\lim_{x\to -1} g(x)$ exist?
(b) Find $\lim_{x\to 1} g(x)$ and $g(1)$.
(c) Describe the behavior of g near $x = 3$ with limit notation.
(d) Find $\lim_{x\to 0} g\big(g(x)\big)$.

Solution

(a) 2, 0, 3; no (one-sided limits differ). (b) 2; $g(1) = 4$. (c) $\lim_{x\to 3^-} g(x) = +\infty$, $\lim_{x\to 3^+} g(x) = -\infty$. (d) $g(x) \to 1$ without equaling 1, and $g(u) \to 2$ as $u \to 1$: the limit is 2.